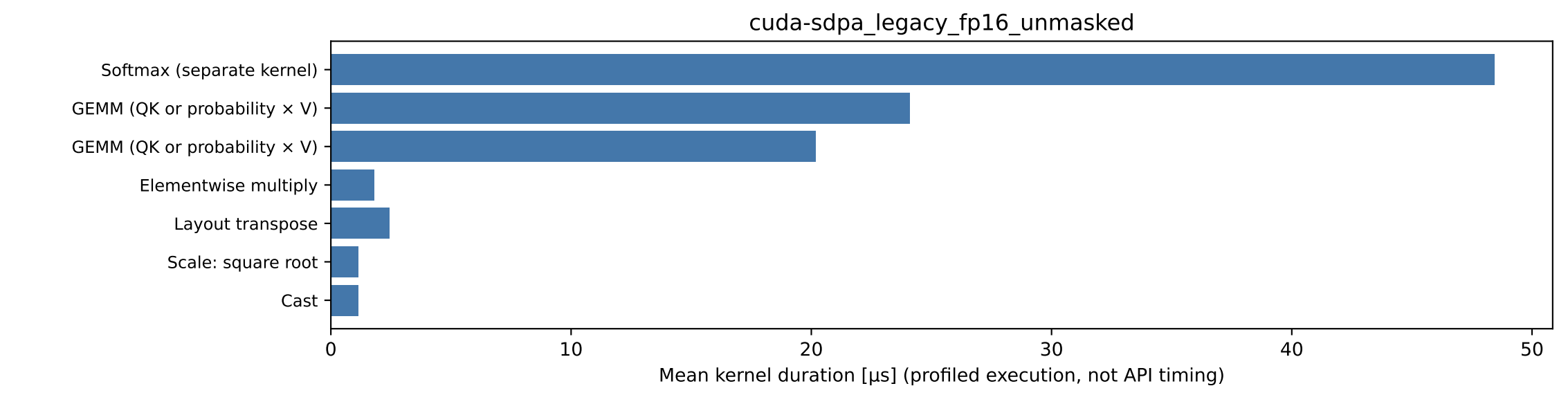
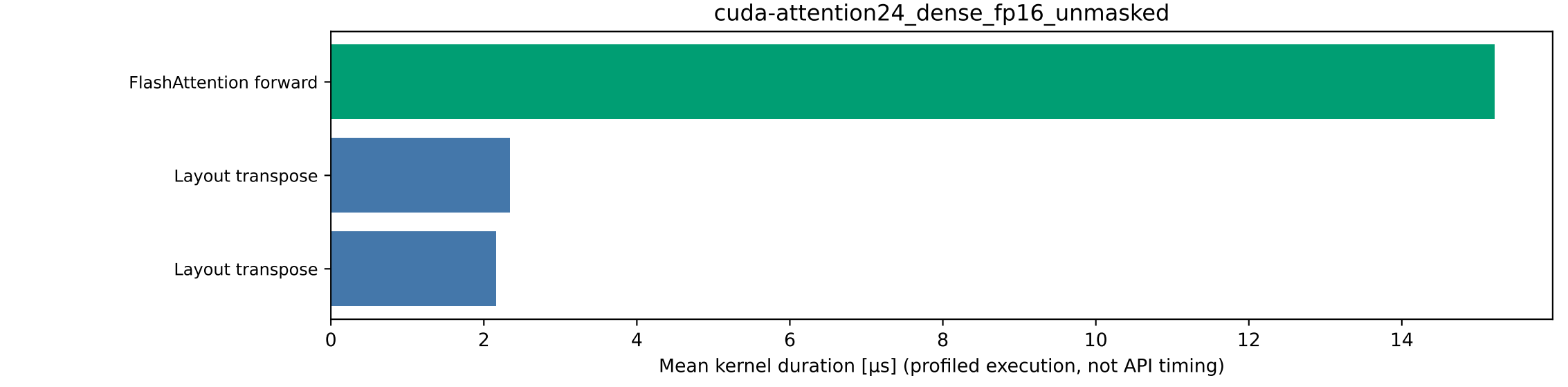
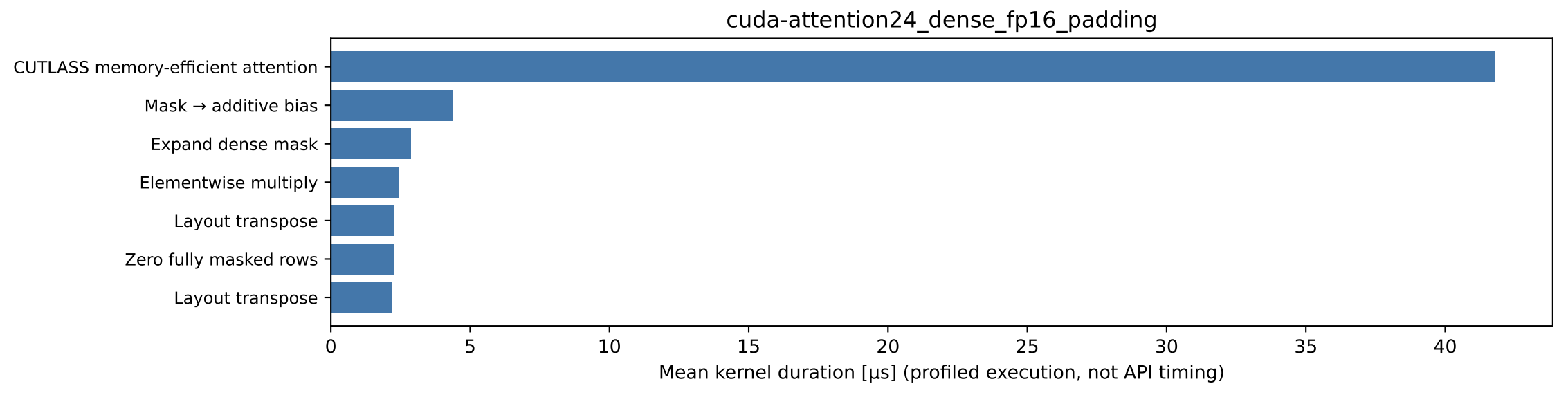
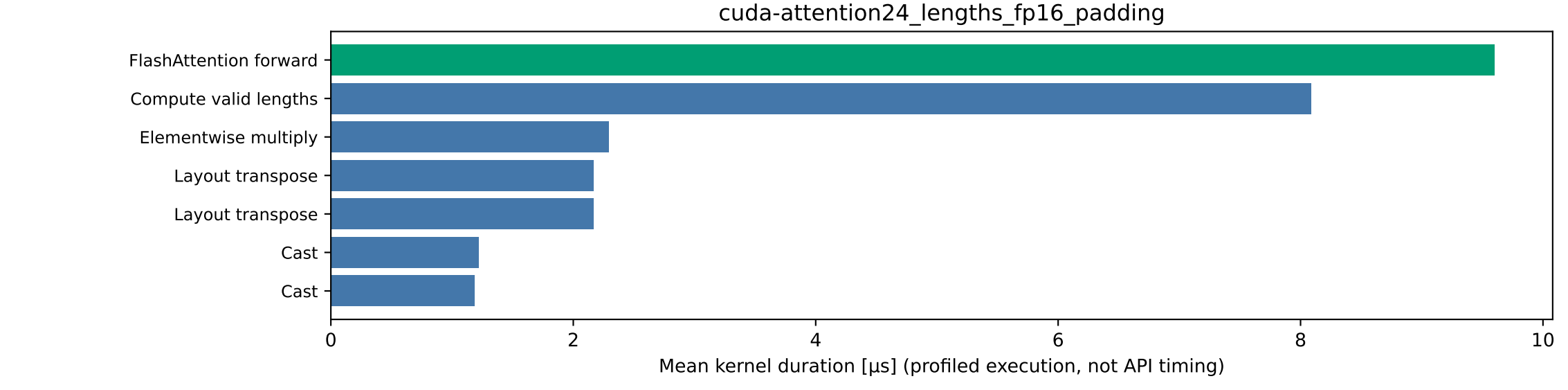

cuda-sdpa_legacy_fp16_unmasked
Softmax (separate kernel)
GEMM (QK or probability × V)
GEMM (QK or probability × V)
Elementwise multiply
Layout transpose
Scale: square root
Cast
0
10
20
30
40
50
Mean kernel duration [µs] (profiled execution, not API timing)


cuda-attention24_dense_fp16_unmasked
FlashAttention forward
Layout transpose
Layout transpose
0
2
4
6
8
10
12
14
Mean kernel duration [µs] (profiled execution, not API timing)


cuda-attention24_dense_fp16_padding
CUTLASS memory-efficient attention
Mask → additive bias
Expand dense mask
Elementwise multiply
Layout transpose
Zero fully masked rows
Layout transpose
0
5
10
15
20
25
30
35
40
Mean kernel duration [µs] (profiled execution, not API timing)


cuda-attention24_lengths_fp16_padding
FlashAttention forward
Compute valid lengths
Elementwise multiply
Layout transpose
Layout transpose
Cast
Cast
0
2
4
6
8
10
Mean kernel duration [µs] (profiled execution, not API timing)